

SNA lab3

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2026-02-15

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[1] 29

gender
0 1
15 14

gender_lab
boy girl
14 15

[1] "friends" "gender" "gender_lab" "liking" "popular"
[6] "student_vars"

Introduction

This assignment investigates hierarchical processes in adolescent peer networks using exponential random graph models (ERGMs). The theoretical motivation is rooted in research on peer status differentiation during adolescence, where two related but distinct forms of social standing are identified: acceptance, typically measured through liking nominations, and popularity, measured through recognition of visibility and prestige. Although these constructs are related, they capture different dimensions of social hierarchy and may be driven by different network mechanisms.

Building on classical theories such as the Matthew effect, preferential attachment, and sociodynamic processes, this assignment explores whether hierarchical tendencies are present in three types of networks: friendship, liking nominations, and popularity nominations. In particular, the focus lies on understanding what determines high indegree in these networks and whether mechanisms such as preferential attachment operate differently across them.

To address these questions, we combine conceptual reasoning with empirical network modelling. First, theoretical expectations are formulated regarding hierarchy and status differentiation in different types of peer relations. Second, descriptive network measures are compared across networks. Third, exponential random graph models are estimated to evaluate whether mechanisms such as reciprocation, gender homophily, transitive closure, and preferential attachment help explain the observed network structures. Finally, model comparison and goodness-of-fit assessment are used to evaluate how well these mechanisms capture the structure of adolescent peer hierarchies.

1. Explain your expectation!

I would expect preferential attachment, or the Matthew effect, to be stronger in the popularity network than in the liking network. Preferential attachment refers to a cumulative advantage process in which individuals who already receive many nominations are more likely to receive additional ones. In social terms, this implies that status reinforces itself: those who are already seen as important, visible, or prestigious attract even more recognition. Popularity nominations are conceptually closer to status attribution than liking nominations. When students nominate someone as popular, they are not necessarily expressing a personal emotional preference, but rather recognizing a perceived social standing. Such reputational processes are highly visible and socially shared. If many peers consider someone popular, others are more likely to do the same, either because they genuinely perceive high status or because popularity becomes socially self-reinforcing. This

creates a classic cumulative advantage dynamic. In contrast, liking nominations reflect personal affect or friendship preference. While popular individuals may also be liked, liking is more relational and embedded in local interactions. It is influenced by reciprocity, shared activities, and homophily rather than purely by reputational prestige. Therefore, although some preferential attachment could still occur in liking networks, it is expected to be weaker.

2. Friendship expectation

If friendship contains elements of both liking and popularity, then we would indeed expect the strength of the Matthew effect in the friendship network to lie somewhere between the two. Friendship nominations typically reflect mutual interaction, emotional closeness, and ongoing social exchange, which makes them similar to liking. At the same time, friendship can also be influenced by status considerations. Highly visible or socially central individuals may attract more friendship nominations simply because they are widely known, socially active, or embedded in multiple groups. In this sense, friendship can contain a reputational component similar to popularity. However, friendship is usually more demanding than simply liking or perceiving someone as popular. It often implies reciprocity, shared time, and stronger relational commitment. These structural features tend to limit extreme cumulative advantage processes. While popular students might receive many popularity nominations, and liked students may receive many liking nominations, friendship ties are constrained by time, emotional investment, and network closure. This can dampen pure preferential attachment dynamics. We would expect the Matthew effect to be strongest in the popularity network, weakest in the liking network, and of intermediate strength in the friendship network. Friendship combines status-driven visibility with relational constraints, producing a cumulative advantage process that exists, but is less extreme than in pure popularity nominations.

3. Control mechanisms

It is a good idea to include gender homophily, reciprocation, and transitive closure when analyzing adolescent networks, even if the main focus is on hierarchy, because these mechanisms represent fundamental structural processes that shape peer relations independently of status differentiation. First, gender homophily is one of the most robust and well-documented patterns in adolescent networks. Young people tend to interact more frequently with same-gender peers, especially during early and middle adolescence. If gender homophily is not included in the model, patterns of clustering or uneven indegree might be mistakenly attributed to hierarchy when they are in fact driven by gender-based segmentation. Including homophily

helps control for similarity-based tie formation and prevents biased interpretation of status effects. Second, reciprocation captures the tendency for ties to be mutual. In friendship and liking networks especially, receiving a nomination increases the likelihood of returning one. Reciprocity generates structural dependencies in the network that directly influence indegree. Without accounting for reciprocity, high indegree could be misinterpreted as hierarchical prominence, when in reality it may reflect balanced mutual exchanges. Including reciprocation allows us to distinguish hierarchical asymmetry from reciprocal bonding. Third, transitive closure represents the tendency for friends of friends to become connected. Adolescents often form tightly knit groups, and these closure processes produce clustering and local cohesion. Transitivity can indirectly affect hierarchical structures because individuals embedded in dense clusters may accumulate nominations simply due to structural position rather than prestige. If transitive closure is not modeled, hierarchical effects may be overstated.

4. Hierarchy index

A good index to express hierarchy, or hub formation, in terms of indegree is indegree centralization.

Indegree centralization summarizes how unequal the indegree distribution is across individuals, relative to a theoretical maximum for a network of the same size. It answers the question: “To what extent is this network organized around one, or a few, highly nominated individuals?” The logic is that a strongly hierarchical network has a “top” structure, where a small number of nodes receive many nominations and most others receive few. Indegree centralization captures exactly this kind of differentiation. It is especially appropriate for adolescent popularity or status networks, where hierarchy often takes the form of a small elite receiving a disproportionate share of nominations. This index is also attractive because it is interpretable and comparable across networks: if we compute indegree centralization for the friendship, liking, and popularity networks, higher values indicate stronger hierarchy or hub formation. In contrast, lower values indicate a more equal distribution of nominations, suggesting weaker hierarchy.

5. Descriptive comparison

```
# A tibble: 3 x 6
  network    density reciprocity transitivity gender_homophily
  <chr>      <dbl>      <dbl>      <dbl>      <dbl>
1 friendship 0.195        0.443        0.497        0.504
2 liking     0.330        0.373        0.636        0.126
```

```
3 popularity 0.0406      0      0.219      0.241
# i 1 more variable: indegree_centralization <dbl>
```

Table 1: Comparison of network properties across the three directed nomination networks.

Network	Density	Reciprocity	Transitivity	Gender	
				homophily	Indegree centralization
friendship	0.195	0.443	0.497	0.504	0.127
liking	0.330	0.373	0.636	0.126	0.134
popularity	0.041	0.000	0.219	0.241	0.138

In Table 1, I compare the three directed nomination networks, friendship, liking, and popularity, in terms of density, reciprocity, transitivity, gender homophily, and my hierarchy index from Exercise 4, indegree centralization. Density indicates how common nominations are in each network. The liking network is the most dense, with a value of 0.330, followed by friendship with 0.195, while popularity is much sparser at 0.041. This suggests that liking nominations are relatively frequent, friendship nominations are more selective, and popularity nominations are the most restrictive form of recognition in this group. Reciprocity is relatively high in friendship, 0.443, and liking, 0.373, indicating a substantial tendency for mutual nominations in these networks. In contrast, reciprocity in the popularity network is 0.000. This is consistent with the interpretation of popularity as a hierarchical status construct rather than a mutual relationship. Being recognized as popular does not require reciprocal acknowledgment.

Transitivity is highest in the liking network, 0.636, and also substantial in friendship, 0.497, while much lower in popularity, 0.219. This means that liking and friendship networks show stronger tendencies toward triadic closure, where friends or liked peers of friends are also connected. Popularity appears less structured by cohesive group processes and more by individual visibility.

Gender homophily is strongest in the friendship network, 0.504, moderate in popularity, 0.241, and lowest in liking, 0.126. This suggests that friendships are more segregated by gender, while liking nominations are more likely to cross gender boundaries. Popularity again lies in between, possibly reflecting both social grouping and broader recognition processes.

Finally, using indegree centralization as my hierarchy index from Exercise 4, I find relatively similar values across networks: 0.127 for friendship, 0.134 for liking, and 0.138 for popularity. Since indegree centralization captures how unequal the distribution of nominations is, these results indicate that all three networks show some degree of differentiation around highly nominated individuals. Popularity

is slightly more centralized, which aligns with the idea that status hierarchies are organized around a small elite. However, the differences are modest, suggesting that hierarchy is present in all three networks rather than being exclusive to popularity.

6. fit an exponential random graph

Term	Friendship	Liking	Popularity
Edges	-3.705 (0.353)	51674.561 (10.555)	-2.970 (0.430)
Mutual	0.201 (0.287)	75.529 (0.273)	fixed at -Inf
Gender homophily (nodematch)	0.314 (0.074)	-37.820 (0.089)	0.579 (0.304)
Transitive closure (GWESP 0.7)	1.738 (0.131)	2.330 (0.188)	1.876 (0.440)
Two-path	-0.169 (0.031)	16.974 (0.085)	-0.304 (0.150)
Preferential attachment (GW indegree 0.7)	3.271 (1.556)	fixed at -Inf	-1.102 (0.640)

Across the three directed nomination networks, the ERGM results reported in Table ?@tbl-q6-ergm-comparison show both common structural tendencies and important differences in magnitude and stability. Because some models exhibited degeneracy and mixing problems, especially in the liking and popularity networks, we followed the professor’s instructions and estimated the models using the Stochastic Approximation algorithm with a reduced initial gain parameter to improve stability. Even so, some coefficients remain unstable or fixed, and these must be interpreted with caution. First, the edges term is negative in all three networks, indicating overall sparsity. The estimates are -3.705 (0.353) for friendship, -51674.561 (10.555) for liking, and -2.970 (0.430) for popularity. For friendship and popularity, the negative coefficients clearly reflect low baseline tie probabilities. The extremely large negative estimate in the liking network, however, indicates estimation instability rather than a substantively meaningful magnitude. Reciprocity (mutual) varies strongly across networks. In the friendship network, the estimate is positive but modest, 0.201 (0.287), suggesting limited additional reciprocation once other structural mechanisms are controlled. In the liking network, the mutual parameter is extremely large and positive, 75.529 (0.273), again indicating instability under the required specification. In the popularity network, the mutual parameter is fixed at $-\text{Inf}$, meaning that mutual nominations are at their minimum attainable value in the data. This implies that reciprocity is structurally absent in the popularity network. The model therefore penalizes reciprocated ties infinitely strongly. This result is substantively coherent: popularity nominations represent reputational recognition rather than mutual exchange. Gender homophily (nodematch) is positive in friendship, 0.314 (0.074), and popularity, 0.579 (0.304), indicating that same-gender nominations are more likely than cross-gender nominations in

these networks. In the liking network, the estimate is -37.820 (0.089), which is again a sign of instability rather than clear evidence of reversed homophily. Overall, we find robust evidence of gender segmentation, especially in friendship and popularity, consistent with well-established adolescent homophily patterns. Transitive closure, captured by the GWESP term, is positive in all networks: 1.738 (0.131) for friendship, 2.330 (0.188) for liking, and 1.876 (0.440) for popularity. This indicates strong clustering tendencies across networks. Nominations tend to close triangles rather than remain independent, which aligns with expectations about peer group cohesion. This effect appears relatively stable across the three models. The twopath term is negative in friendship, -0.169 (0.031), and popularity, -0.304 (0.150), helping to stabilize the model by counterbalancing excessive clustering. In the liking network, the positive and large estimate, 16.974 (0.085), again signals instability. As discussed in class, the twopath term is included to reduce degeneracy-related problems, and its presence is methodologically important even when interpretation is limited.

Preferential attachment, measured by the GW indegree term, differs substantially. In the friendship network, the estimate is positive, 3.271 (1.556), suggesting some concentration of incoming nominations on a smaller set of individuals. This provides moderate evidence of a Matthew effect in friendship. In the liking network, the parameter is fixed at $-\text{Inf}$, indicating that cumulative advantage cannot be stably estimated under the imposed specification. In the popularity network, the estimate is -1.102 (0.640), suggesting no clear evidence for preferential attachment once other structural mechanisms are controlled.

Comparing only what can meaningfully be compared across models, we observe robust evidence of transitive closure and gender homophily, especially in friendship and popularity. Reciprocity is present in friendship but structurally absent in popularity. Preferential attachment appears moderate in friendship but is not clearly supported in popularity and is unstable in liking. These results partially align with our expectations: while we anticipated the strongest Matthew effect in popularity, the structural absence of reciprocity and estimation constraints suggest that popularity nominations operate under different structural rules than liking or friendship. As emphasized by the professor, some of these results may reflect unconverged or partially unstable models, and therefore substantive interpretation must remain cautious.

7. interpret these results

In light of our expectations from Exercises 1, 2, and 3, the ERGM results in Table **?@tbl-q6-ergm-comparison** provide mixed support and also highlight important limitations due to estimation instability in parts of the liking and popularity

models. Following the professor’s guidance, we relied on Stochastic Approximation controls to obtain estimates under degeneracy and mixing issues, and we therefore interpret the results cautiously, focusing mainly on patterns that are stable and substantively comparable across networks.

Regarding our main expectation in Exercise 1, we expected preferential attachment to be stronger in popularity than in liking. The results do not clearly support this. In the popularity network, the GW indegree estimate is negative, -1.102 (0.640), which does not indicate the expected cumulative advantage pattern once other dependencies are controlled. In contrast, the liking network shows a non interpretable outcome for GW indegree, reported as fixed at $-\text{Inf}$, together with other extreme coefficients, indicating that preferential attachment is not reliably estimable under the required specification for this dataset. Because the liking model is unstable, we cannot make a clean popularity versus liking comparison for preferential attachment in the way we initially expected.

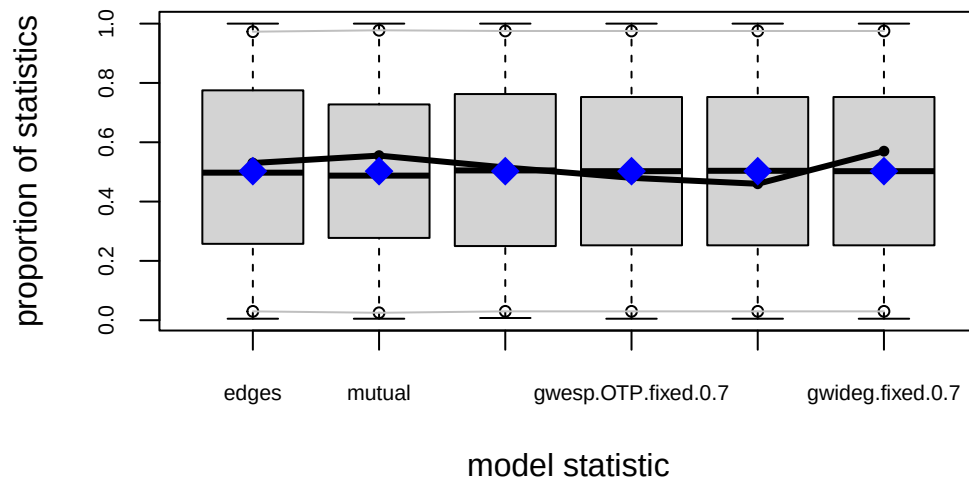
For Exercise 2, we expected the Matthew effect in friendship to be intermediate between liking and popularity. What we can say with confidence is that friendship does show a positive GW indegree effect, 3.271 (1.556), suggesting some concentration of incoming nominations in friendship, consistent with a moderate cumulative advantage dynamic. However, because liking is not stably estimated and popularity shows a negative coefficient, we cannot place friendship “between” the two in a meaningful, comparable way. The most defensible interpretation is simply that friendship displays evidence of indegree concentration, while the other two networks do not provide stable, directly comparable evidence for the same mechanism under this specification.

Exercise 3 focused on the need to control for fundamental structural mechanisms, and here the results align strongly with our expectations. Gender homophily is positive in friendship, 0.314 (0.074), and in popularity, 0.579 (0.304), consistent with the idea that same gender nominations are more likely and that failing to control for this would bias interpretation of other effects. Transitive closure is also consistently positive across networks, with GWESP estimates of 1.738 (0.131) for friendship, 2.330 (0.188) for liking, and 1.876 (0.440) for popularity, supporting our expectation that clustering and local cohesion are central features of adolescent nomination networks. Reciprocity behaves as expected in the sense that it is at least feasible in friendship, 0.201 (0.287), but it is structurally absent in popularity, where the mutual term is fixed at $-\text{Inf}$, indicating that reciprocated popularity nominations essentially do not occur in the observed data. This directly supports our argument that reciprocity must be modelled when present, but also that some nomination types, such as popularity, can be fundamentally asymmetric by construction.

8. optional

9. a brief assessment of goodness of fit

Figure 1: Goodness of fit for the friendship ERGM. The black line is the observed network, boxplots show simulated networks (central 95% intervals).



10. model but only to the friendship network

11. Report converged results

12. collaborated

Table 3

Goodness-of-fit for model statistics

	obs	min	mean	max	MC p-value
edges	158.00000	130.00000	157.17500	187.00000	1.00
mutual	35.00000	23.00000	35.11000	53.00000	1.00
nodematch.gender	119.00000	76.00000	118.23000	155.00000	1.00
gwesp.OTP.fixed.0.7	236.36829	180.94276	234.25565	291.82727	0.96
twopath	816.00000	561.00000	811.87500	1137.00000	0.92
gwideg.fixed.0.7	55.39195	51.98342	55.40035	57.05333	0.86

Goodness-of-fit for minimum geodesic distance

	obs	min	mean	max	MC p-value
1	158	130	157.175	187	1.00
2	244	126	222.405	346	0.56
3	136	44	148.985	239	0.77
4	65	5	73.480	145	0.90
5	14	0	24.920	114	0.86
6	0	0	6.240	52	1.00
7	0	0	1.125	22	1.00
8	0	0	0.135	7	1.00
9	0	0	0.010	1	1.00
Inf	195	28	177.525	440	0.71

Goodness-of-fit for out-degree

	obs	min	mean	max	MC p-value
0	6	1	4.295	9	0.49
1	1	0	2.125	7	0.75
2	0	0	1.395	5	0.44
3	4	0	1.695	7	0.16
4	5	0	2.070	7	0.10
5	3	0	2.600	8	0.91
6	3	0	2.795	12	0.97
7	0	0	2.770	10	0.14
8	2	0	2.825	9	0.93
9	1	0	2.280	7	0.62
10	1	0	1.675	7	0.97
11	0	0	1.060	4	0.60
12	1	0	0.710	4	1.00
13	0	0	0.395	3	1.00
14	0	0	0.185	2	1.00
15	0	0	0.075	1	1.00
16	0	0	0.030	1	1.00
17	1	0	0.010	1	0.02
18	0	0	0.005	1	1.00
19	0	0	0.005	1	1.00
28	1	0	0.000	0	0.00

Goodness-of-fit for in-degree

Figure 2: Goodness of fit for the friendship ERGM. The black line is the observed network, boxplots show simulated networks (central 95% intervals).

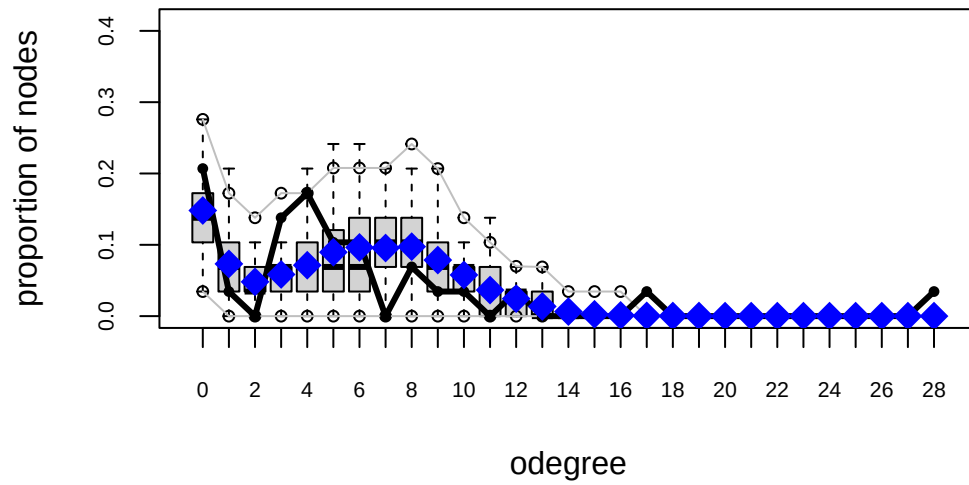


Figure 3: Goodness of fit for the friendship ERGM. The black line is the observed network, boxplots show simulated networks (central 95% intervals).

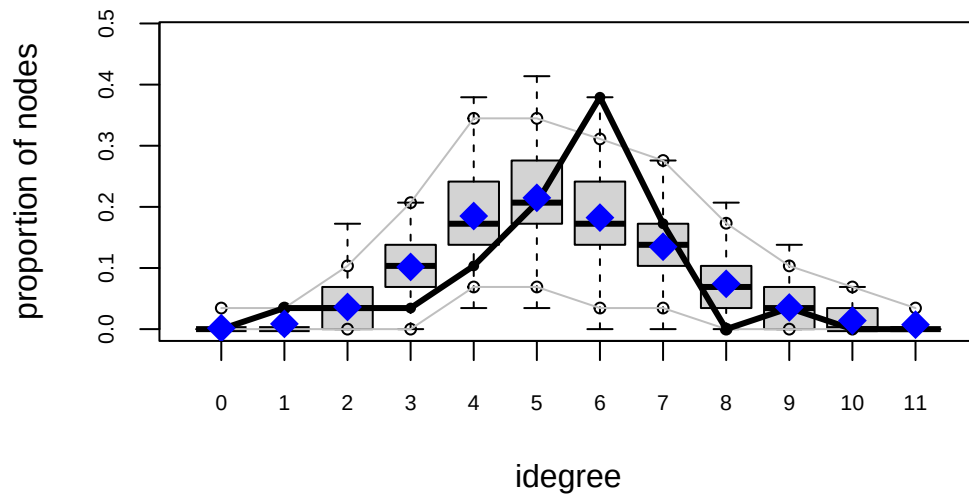


Figure 4: Goodness of fit for the friendship ERGM. The black line is the observed network, boxplots show simulated networks (central 95% intervals).

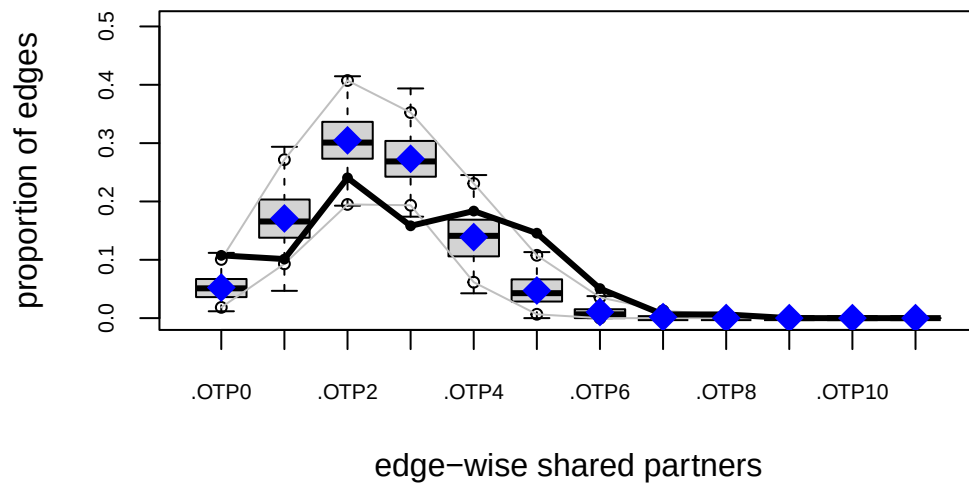


Figure 5: Goodness of fit for the friendship ERGM. The black line is the observed network, boxplots show simulated networks (central 95% intervals).

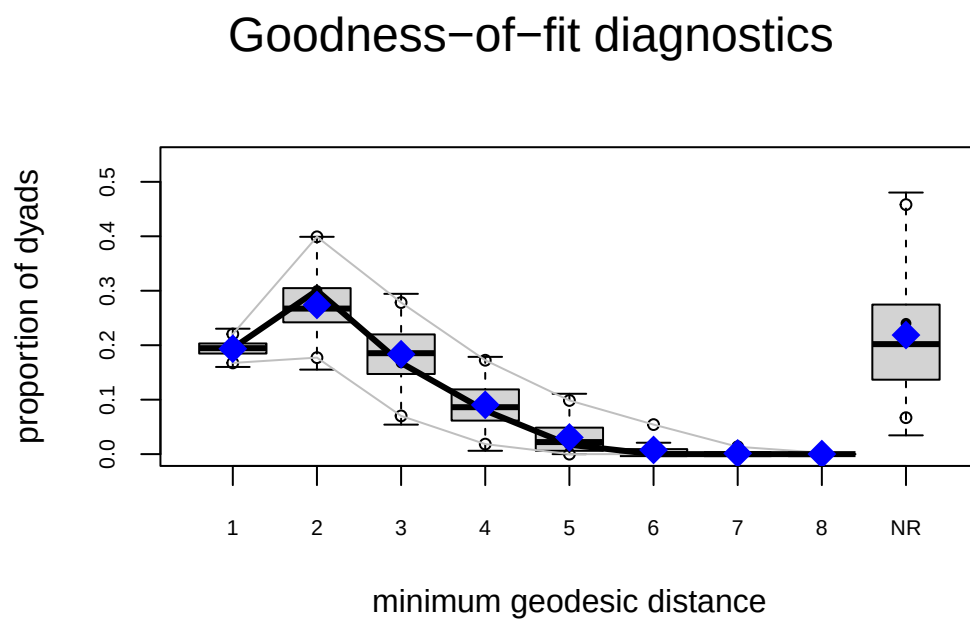


Figure 6: Goodness of fit for the liking ERGM. Interpretation should be cautious due to estimation instability noted in Exercise 6.

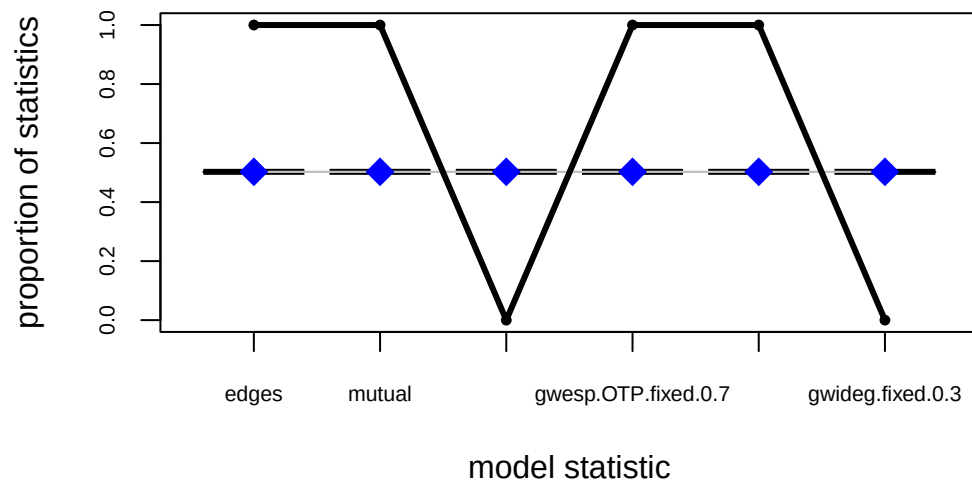


Figure 7: Goodness of fit for the liking ERGM. Interpretation should be cautious due to estimation instability noted in Exercise 6.

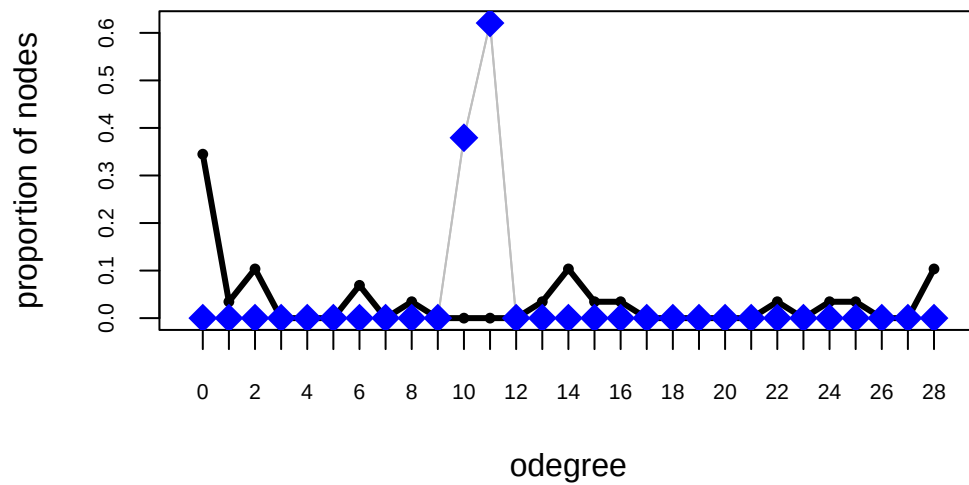


Figure 8: Goodness of fit for the liking ERGM. Interpretation should be cautious due to estimation instability noted in Exercise 6.

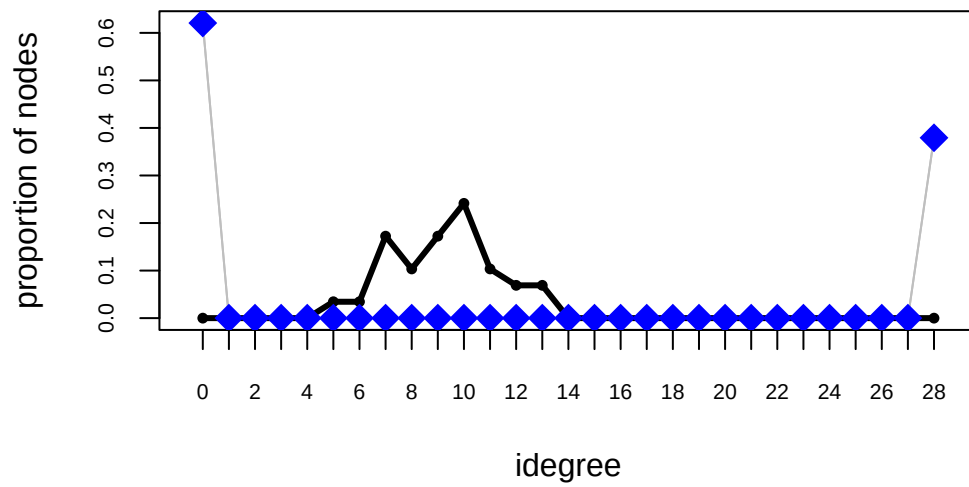


Figure 9: Goodness of fit for the liking ERGM. Interpretation should be cautious due to estimation instability noted in Exercise 6.

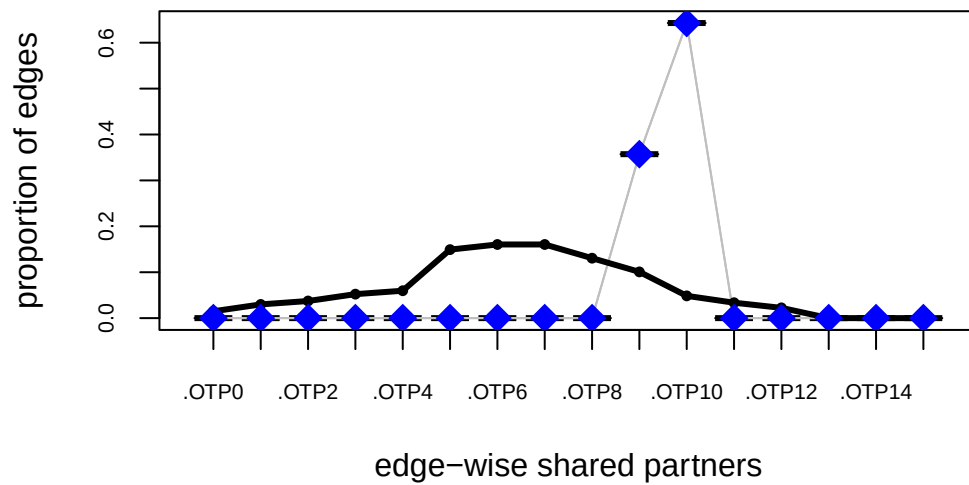


Figure 10: Goodness of fit for the liking ERGM. Interpretation should be cautious due to estimation instability noted in Exercise 6.

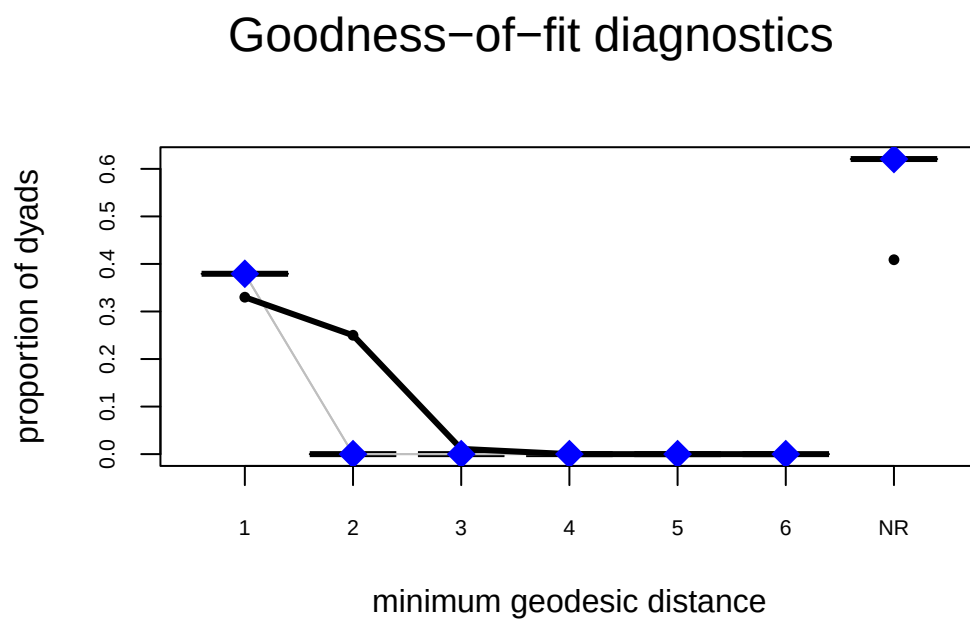


Figure 11: Goodness of fit for the popularity ERGM. Interpretation should be cautious due to structural constraints and fixed parameters in Exercise 6.

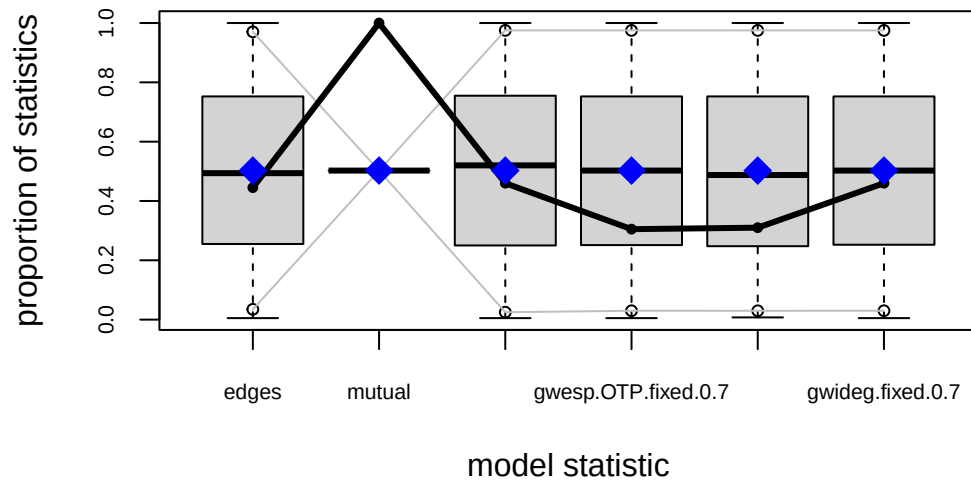


Figure 12: Goodness of fit for the popularity ERGM. Interpretation should be cautious due to structural constraints and fixed parameters in Exercise 6.

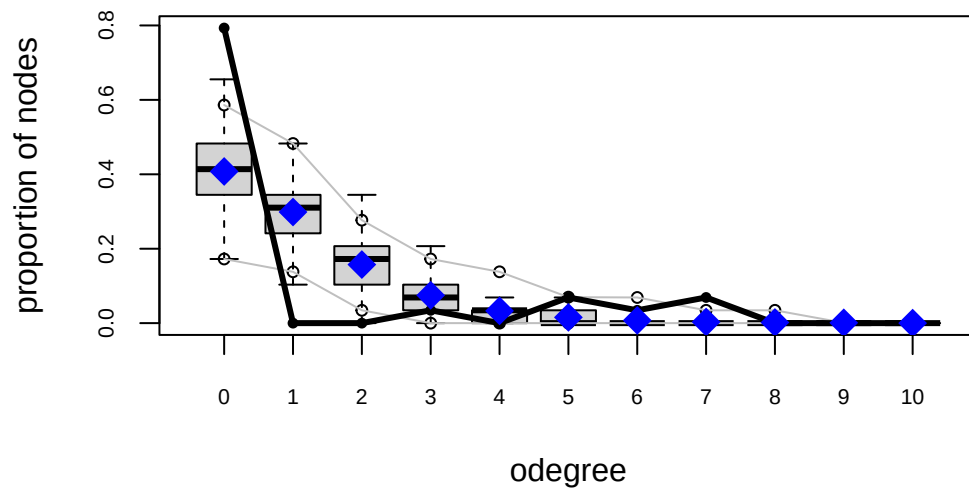


Figure 13: Goodness of fit for the popularity ERGM. Interpretation should be cautious due to structural constraints and fixed parameters in Exercise 6.

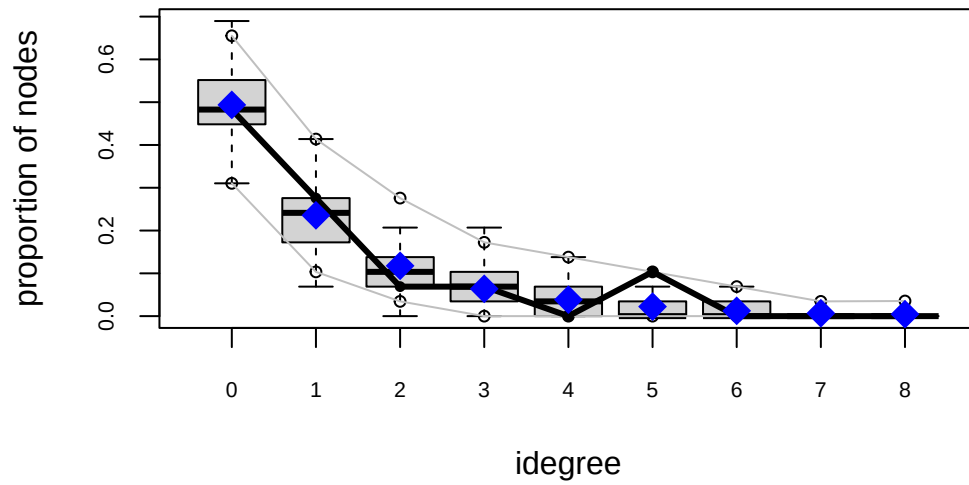


Figure 14: Goodness of fit for the popularity ERGM. Interpretation should be cautious due to structural constraints and fixed parameters in Exercise 6.

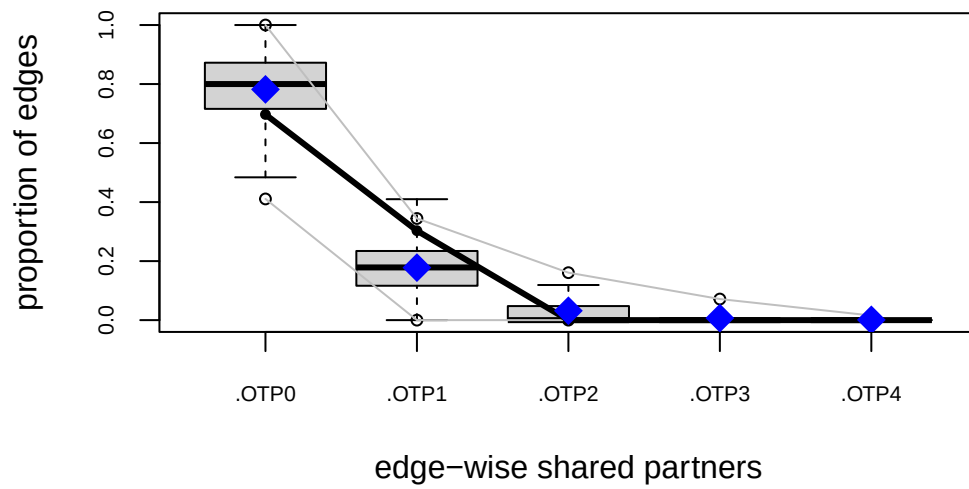


Figure 15: Goodness of fit for the popularity ERGM. Interpretation should be cautious due to structural constraints and fixed parameters in Exercise 6.

